

Geethanjali College of Engineering and Technology (Autonomous)

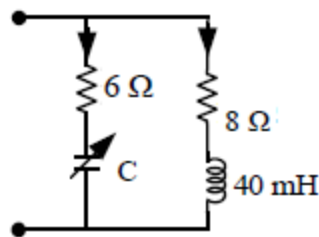
II B. Tech. EEE - I Semester

ELECTRICAL CIRCUIT ANALYSIS (20EE21003)

ASSIGNMENT-II

UNIT-III

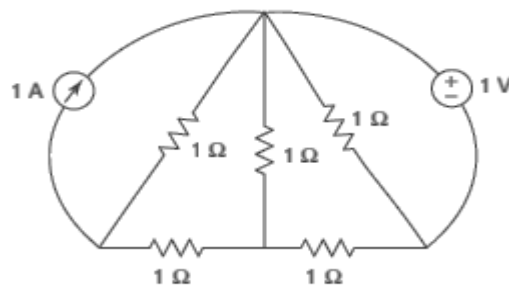
- 1 Obtain the current locus of a series RL circuit.
- 2 Obtain the current locus of a series RC circuit.
- 3 Define Resonance, Band width and Q factor. Write formulas
- 4 Find the value of C for which the circuit as shown in Fig. is resonance at 3000 rad/sec.



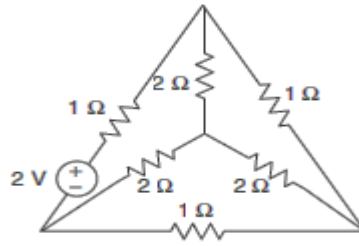
- 5 Derive the expression for resonant frequency in parallel resonant circuit.
- 6 Obtain the Expression for the Bandwidth of a series resonant circuit.
- 7 A series RLC circuit of $R = 100\ \Omega$, $L = 0.01\text{ H}$ and $C = 0.01\ \mu\text{F}$ is connected Across a single phase ac supply. Determine i) f_0 ii) Q factor, iii) BW iv) f_1 and f_2
- 8 It is required that a series RLC circuit should resonate at 500 KHz. Determine the values of R, L, and C if the bandwidth of the circuit is 10 KHz and its impedance is $100\ \Omega$ at resonance. Also find the voltages across L and C resonance if the applied voltage is 75V.

UNIT-IV

- 1 For the network shown in Fig. draw the oriented graph. Also draw the no. of possible trees and selecting any one tree Write the cut set schedule.



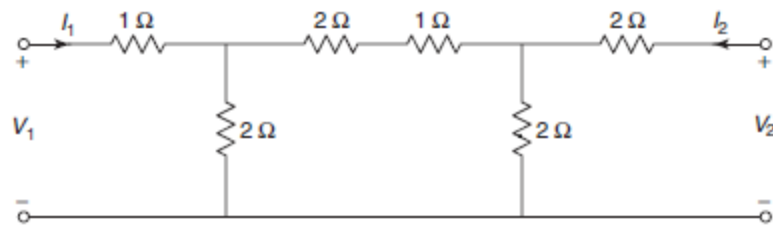
- 2 For this network shown in Fig., write down the tie set matrix and obtain the network equilibrium equation in matrix form using KVL. Calculate the loop currents and branch currents.



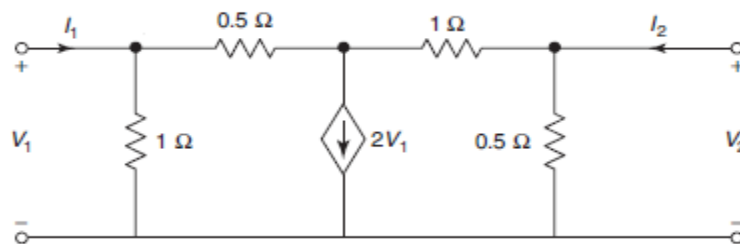
- 3 Define Duality. Explain procedure to draw the dual network.
- 4 Define the following terms.
 - i) Tree. ii) Co-Tree iii) Branch iv) Link
 - v) Twig vi) Node
- 5 The following direct current measurement were made on the two port network: Find h parameters

Port 1 open	Port 2 short circuited
$V_1 = 2 \text{ V}$	$I_1 = -5 \text{ mA}$
$V_2 = 10 \text{ V}$	$I_2 = 20 \text{ mA}$
$I_2 = 10 \text{ mA}$	$V_1 = 5 \text{ V}$

- 6 Obtain the relationship Y - Parameters in terms of other two port network parameters.
- 7 Find the Y-parameters for the circuit shown in figure



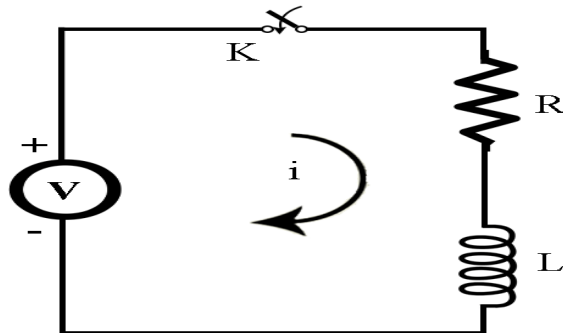
- 8 Write the condition for symmetry and reciprocity for ABCD-parameters.
- 9 Derive the condition for Symmetry for all two port network parameters.
- 10 Determine Z and Y-parameters of the network shown in Fig.



UNIT-V

- 1 Obtain the Transient Response in Series RL and RC circuit with Sinusoidal Excitation.
- 2 Drive the expression for the current in series RL and RC circuits with source free DC excitation
- 3 Write a note on initial conditions of basic circuit elements

- 4 In the circuit shown in the following figure, initially switch 'K' is kept open for long time. At $t=0$ switch K is closed obtain the expression for current in the circuit for $t \geq 0$. Find the value of current in the circuit in one-time constant period? Determine the instant of time at which the current in the circuit reaches to 1.2 A



- 5 Derive the expression for the current in series RL circuit with DC excitation.
- 6 Derive the expression for the current in series RC circuit with DC excitation.
- 7 Draw the m-derived low pass and high pass filter T and Π networks.
- 8 Design a constant-k low-pass T and Π section filters having cut-off frequency of 2 kHz and nominal characteristic impedance of 400Ω .
- 9 Obtain the nominal impedance value of T and Π -networks.
- 10 Explain classification of filters.